

PALGUN THATIPAMULA

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Research Profile

Project Associate at IIT Kharagpur currently working on **molecular communication, Internet of NanoBio Things, and targeted drug delivery**. Current work involves exploring computational and simulation-based approaches for molecular transport and digital-twin modeling. M.Tech. in Artificial Intelligence from DIAT-DRDO with prior research experience in spatiotemporal learning, computer vision, anomaly detection, explainable AI, and simulation of dynamic systems.

Research Experience

Indian Institute of Technology Kharagpur (IIT Kharagpur)

Project Associate — Molecular Communication & NanoBio Systems

Kharagpur, India

Sep 2026 – Present

- Working on the project “*System Design and Analysis of Molecular Communication based Internet of NanoBio Things for Targeted Drug Delivery*”.
- Currently exploring computational models of molecular communication, particle transport, and targeted drug-delivery mechanisms as part of the initial development of the project.
- Developing an initial **3D Brownian-motion and Monte Carlo simulation framework** for studying stochastic molecular trajectories and transport behavior.
- Exploring **digital-twin approaches** for representing molecular transport and targeted drug-delivery systems through simulation-based analysis.

Defence Institute of Advanced Technology (DIAT-DRDO)

Student Intern — MeitY Sponsored Research Project

Pune, India

Sep 2025 – Jun 2026

- Contributed to the *ExxS* Explainable AI research initiative, studying interpretable deep-learning methods for mission-critical systems.
- Investigated feature attribution and model-interpretability techniques to improve transparency and reliability of data-driven decision systems.

Selected Research Work

Molecular Transport Simulation

Python, NumPy, SciPy, Monte Carlo

- Implemented an initial three-dimensional Brownian-motion simulation for studying stochastic particle transport and molecular trajectories.
- Developed Monte Carlo-based analysis of particle propagation and arrival behavior as an initial computational model for molecular communication research.

Interpretable Spatiotemporal Scientific Machine Learning

DenseNet, BiLSTM, Attention, Grad-CAM

- Co-authored *Astro-DBA*, an interpretable dual-view DenseNet-BiLSTM architecture for exoplanet transit detection from photometric time-series observations.
- Applied temporal modeling and Grad-CAM-based attribution to examine whether the model focuses on physically meaningful transit structures.

Real-Time Spatiotemporal Anomaly Detection

YOLOv8, ConvLSTM, Optical Flow

- Developed a cascaded detection, tracking, and temporal anomaly-detection pipeline for UAV behavior analysis.
- Achieved **0.944 AUROC** using YOLOv8-based perception combined with ConvLSTM-based spatiotemporal modeling.

Education

Defence Institute of Advanced Technology (DIAT-DRDO)

M.Tech. in Artificial Intelligence (CSE) — CGPA: 8.35/10

Pune, India

2024 – 2026

Rajiv Gandhi University of Knowledge Technologies (RGUKT)

B.Tech. in Electronics & Communication Engineering — CGPA: 7.85/10

Basar, Telangana, India

2020 – 2024

Publications

- Ajay Waghumbare, **Palgun T**, Yogesh Bhore, and Mandira Mondal, “Astro-DBA: An interpretable dual-view DenseNet-BiLSTM architecture for exoplanet transit detection,” *Experimental Astronomy*, vol. 62, art. no. 8, 2026.
- Palgun Thatipamula** (Main Author), “Real-Time Cascaded Pipeline for Drone Detection and Behavior Anomaly Analysis: YOLOv8 Benchmark and ConvLSTM Approach,” *IEEE PuneCon 2025*.

Intellectual Property

- Patent Co-inventor**, AI-based multi-object tracking and performance-analysis system patent, granted May 2026.

Technical Expertise

Current Research Methods: Molecular Communication (ongoing), Molecular Transport, Brownian Motion, Stochastic Modeling, Monte Carlo Simulation, Digital-Twin Modeling, Targeted Drug Delivery.
AI & Computational Methods: Deep Learning, Spatiotemporal Learning, Anomaly Detection, Computer Vision, Optical Flow, Multi-Object Tracking, Sensor Fusion, Explainable AI.
Scientific Computing: Python, NumPy, SciPy, PyTorch, OpenCV, MATLAB, TensorRT, ONNX.
Computational Systems: Physics-Based Modeling, Real-Time Simulation, Digital-Twin Development, CUDA, Linux, NVIDIA Jetson, NVIDIA GPU Computing.

Research Interests

Molecular Communication; Internet of NanoBio Things; Molecular Transport; Biomedical Digital Twins; Targeted Drug Delivery; Stochastic Particle Dynamics; Multi-Scale Biomedical Simulation; AI for Biomedical Systems.

Selected Honors

UGC-NET 2025 (CSE) | GATE 2024 (ECE) | TG PGECET 2024 — Rank 142 (ECE)